



**JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND
TECHNOLOGY (JKUAT)**

**DEPARTMENT OF TELECOMMUNICATION AND
INFORMATION ENGINEERING**

**TITLE: AN IoT BASED LOAD SHEDDING TIME MANAGEMENT
SYSTEM WITH PROGRAMMABLE INTERFACE**

BY:

LEWIS MURIUKI	ENE221-O106/2019
KISERO BENARD	ENE221-O270/2019

SUPERVISOR: DR. STEPHEN KIAMBI

A Final Year Project Proposal Submitted to the Department of
Telecommunication and Information Engineering in partial fulfilment
requirement for the award of a Bachelor of Science in Telecommunication
and Information Engineering.

FEBRUARY 2026

DECLARATION

We hereby declare that the project entitled 'An IoT based load shedding time management system with programmable interface' submitted to the Department of Telecommunication and Information Engineering, JKUAT, is a record of original work done by us, under the guidance of Dr. Stephen Kiambi.

We solemnly declare to the best of our knowledge that this project report is our original work, except where due acknowledgment is made in the text and has not been previously submitted to Jomo Kenyatta University of Agriculture and Technology or any other institution for the award of a degree or diploma.

NAME: KISERO BENRAD OYOO

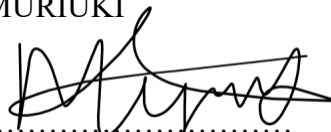
REG. No.: ENE221-0270/2019

SIGNATURE: 

DATE: 26th February, 2026

NAME: LEWIS MURIUKI

REG. No.: ENE221-0106/2019

SIGNATURE: 

DATE: 26th February, 2026

SUPERVISOR CONFIRMATION

This is to certify that the project entitled 'An IoT based load shedding time management system with programmable interface' carried out by the above-named students, has been read and approved for meeting part of the requirements and regulations governing the award of Bachelor of Science in Telecommunication and Information Engineering of JKUAT, Kenya.

NAME: DR. STEPHEN KIAMBI

SIGNATURE: 

DATE: **26 Feb. 2026**

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ABSTRACT

This project is motivated by practical industrial intensive research on challenges facing consumer Kenya Power and Lighting Company, where firsthand exposure to real-world power distribution operations revealed significant challenges associated with load management and outage control. During the research period, it was observed that load shedding processes are largely manual, reactive, and lack intelligent prioritization, resulting in inefficient power distribution, prolonged outages, and minimal real-time feedback to both operators and consumers. These limitations highlighted the need for a more adaptive, automated, and data-driven approach to managing power demand, especially during peak load conditions, planned maintenance, and unexpected system faults. The proposed system integrates IoT-enabled sensors and smart meters to continuously collect real-time data on power consumption, voltage levels, and network load conditions across different zones. This data is transmitted to a central control platform, where a programmable software interface allows system operators to define load priorities, scheduling policies, and operational constraints based on critical infrastructure requirements such as hospitals, industries, and residential areas. Finally, the proposed solution aims to enhance the reliability, efficiency, and intelligence of power distribution systems by enabling predictive and responsive load management. By providing a programmable, scalable, and real-time control framework, the system offers a practical technological solution that can support modern smart grid initiatives, improve customer satisfaction, and significantly reduce the negative economic and social impacts associated with poorly managed load shedding.

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LIST OF ABBREVIATIONS

IoT- Internet of Things

IC- Integrated Circuits

IDE- Integrated Development Environment

PCB- Printed Circuit Board

LCD- Liquid Crystal Display

AC- Alternating Current

DC- Direct Current

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CHAPTER ONE

1. INTRODUCTION

1.1 Background Information

The rapid growth in electricity demand driven by population increase, urbanization, industrial expansion, and the widespread use of electrical appliances has placed significant strain on power generation and distribution systems, particularly in developing countries. Power utilities are increasingly faced with challenges such as limited generation capacity, aging infrastructure, transmission losses, and frequent system faults, which often result in power imbalances between supply and demand. When demand exceeds available supply, utilities are forced to implement load shedding as a necessary measure to prevent total system collapse and maintain grid stability.

In Kenya, electricity distribution and retail services are primarily managed by the Kenya Power and Lighting Company, which serves millions of customers across residential, commercial, and industrial sectors. Despite ongoing investments in power generation and grid expansion, the country continues to experience periodic power shortages, planned outages, and unplanned blackouts due to maintenance activities, equipment failures, and peak demand conditions. These outages often have serious economic and social consequences, including reduced productivity, disruption of essential services such as healthcare and communication, and customer dissatisfaction.

Traditionally, load shedding in many power distribution systems is conducted using manual or semi-automated methods, where operators rely on fixed schedules, telephone communication, and limited real-time monitoring tools. This approach lacks intelligence, flexibility, and responsiveness, making it difficult to prioritize critical loads, adapt to real-time demand changes, or optimize outage durations. As a result, load shedding is often inefficient, unfairly distributed, and prone to human error, leading to prolonged outages and ineffective utilization of available power resources.

With the emergence of smart grid technologies and the Internet of Things (IoT), new opportunities have arisen for improving power system monitoring, control, and automation. IoT enables the deployment of interconnected sensors, smart meters, and embedded devices that can collect and transmit real-time data on power usage, system health, and network conditions. This technological shift makes it possible to design intelligent load management systems that are data-driven, programmable, and capable of making autonomous decisions.

Therefore, there is a growing need for an intelligent and automated load shedding framework that can dynamically manage power distribution based on real-time system conditions and predefined priorities. This project, titled “An IoT-Based Load Shedding Time Management System with Programmable Interface”, is developed within this context to address the limitations of traditional load shedding approaches and contribute towards the realization of efficient, reliable, and sustainable smart power distribution systems.

1.2 Problem Statement

In many developing power distribution systems, load shedding is still implemented using manual or semi-automated methods that rely on fixed schedules, limited system monitoring, and human intervention. These traditional approaches lack real-time visibility into network conditions and do not support intelligent decision-making based on actual demand patterns or critical infrastructure requirements.

As a result, load shedding operations are often inefficient, inflexible, and poorly optimized, leading to prolonged and uneven power outages, underutilization of available power resources, and minimal prioritization of essential services such as hospitals, communication systems, and industrial facilities. Furthermore, the absence of programmable control interfaces makes it difficult for system operators to dynamically adjust load shedding policies or respond promptly to changing network conditions. These limitations reduce overall system reliability, increase operational costs, and negatively impact economic productivity and customer satisfaction.

1.3 Proposed Solution

This project proposes an IoT-Based Load Shedding Time Management System with a Programmable Interface that enables intelligent, automated, and real-time control of power distribution. The proposed system integrates IoT-enabled sensors and smart meters to

continuously monitor electrical parameters such as power consumption, voltage levels, current flow, and load demand across different distribution zones. This real-time data is transmitted to a central control platform where it is analyzed to determine system conditions and identify potential overload situations.

The system incorporates a programmable software interface that allows power system operators to define load priorities, scheduling rules, and operational constraints based on the importance of different consumer categories, including critical infrastructure, commercial users, and residential areas. Using this interface, operators can configure policies that automatically regulate load shedding actions without requiring continuous manual intervention.

In addition, microcontrollers and automated switching devices are used to physically control load connections, enabling the system to dynamically disconnect or reconnect loads based on predefined rules and real-time network conditions. The proposed solution contributes towards the development of a smarter, more efficient, and responsive power distribution system aligned with modern smart grid technologies.

1.4 OBJECTIVES

1.4.1 Main Objective

To design an IoT-enabled load control system capable of remotely managing electrical loads using wireless communication technologies.

1.4.2 Specific Objective

To develop a programmable interface that allows authorized users to set load shedding schedules and assign priority levels to different loads.

To implement automated timing and control mechanisms that execute load shedding based on predefined rules and real-time system conditions.

To enable real-time monitoring of load status, system performance, and power availability through a web or mobile-based interface.

CHAPTER TWO

2. LITERATURE REVIEW

2.1 Introduction

Most existing load shedding solutions are either manually controlled, rigidly scheduled, or designed as part of large-scale smart grid infrastructures that are costly and complex to implement for KENYA POWER AND LIGHTING COMPANY (KPLC). Some of the proposed solutions we found have particular flaws, the majority of these solutions focus on basic remote switching, fixed time schedules, or general energy management rather than providing a flexible, user-programmable, and adaptive load shedding time management framework.

The proposed IoT-Based Load Shedding Time Management System with a Programmable Interface is fundamentally different from conventional approaches in both design and functionality. Unlike traditional microcontroller-based load shedding systems that rely logic and static schedules, this project introduces a programmable interface that allows users to define, modify, and prioritize load shedding rules in real time. This shifts control from fixed embedded logic to an intelligent, user-centric system capable of responding to varying power conditions and user requirements.

Furthermore, while many existing IoT load management systems emphasize simple ON/OFF control through mobile applications, they often lack advanced scheduling flexibility, load prioritization, and adaptive decision-making based on real-time conditions [1]. The proposed system bridges this gap by integrating IoT connectivity with programmable time management and conditional control logic, enabling intelligent load shedding decisions rather than remote operation. This approach allows the system to operate not only as a monitoring tool but as an active decision-making platform.

By combining real-time sensing, IoT communication, and a programmable user interface, this project contributes a newer and practical solution that lies between basic automated load shedding and complex smart grid systems [2]. It is designed to be scalable, cost-effective, and adaptable to real-world load shedding scenarios, making it particularly suitable for deployment in environments where flexibility, reliability, and user control are critical. As a result, this project offers a clear advancement over existing solutions and demonstrates originality in both its technical implementation and its application focus.

2.2 Related works

We analyzed the existing proposed solutions to load shedding time management systems and did a thorough analysis of the ones relevant with our project and enough information. Each work is analyzed in terms of system architecture, methodology, findings, strengths, weaknesses, and limitations. The review concludes by identifying research gaps that justify the proposed “IoT-Based Load Shedding Time Management System with Programmable Interface”.

2.2.1 Smart Load Shedding using IoT and Machine Learning [3]

This study proposed a smart load shedding system that uses IoT sensors and machine learning to dynamically predict grid demand and perform load shedding based on real-time data. The system collects voltage/current values from endpoints and uses classification algorithms to decide which loads to disconnect during peak demand.

Load shedding times and control logic are fixed in their microcontroller’s code, so any change requires technical reprogramming of the device. The system uses a mobile or web application, they usually allow only simple actions such as setting fixed ON/OFF times or manually switching loads. They do not allow users to create flexible rules, combine time schedules with real-time conditions, or adjust system behavior as power situations change.

The authors developed a smart load shedding framework motivated by the increasing instability of modern power grids caused by rapid urbanization, variable renewable energy integration, and rising peak demand. Traditional load shedding techniques, which rely on frequency-based relays or fixed time schedules, were identified as inefficient because they disconnect entire feeders without discrimination and cause unnecessary power interruptions.

The authors proposed that integrating IoT-enabled sensing with machine learning analytics could transform load shedding from a static emergency mechanism into an intelligent predictive control system.

2.2.1.1 Key findings

From their research we were able to conclude the following:

Singh and Sharma’s study reported 37% reduction in overload occurrences when compared to conventional fixed-schedule load shedding. This improvement is primarily attributed to the predictive capability of the machine learning model.

The predictive intervention contributed to more stable voltage profiles across distribution nodes. Because overload conditions were mitigated early, the system avoided severe voltage dips.

Reduction of Unnecessary Disconnections- Conventional load shedding disconnects entire regions, including consumers who may not significantly contribute to peak load. The intelligent classification system enabled selective shedding, targeting specific non-essential loads rather than entire feeders.

The system is automated and data-driven, it eliminates delays caused by manual intervention. The machine learning classifier processes real-time data and triggers relays almost immediately when stress patterns are detected.

2.2.1.2 Strengths

Predictive Intelligence Rather than Reactive Control

The most powerful strength of this study lies in its shift from reactive to predictive control. Reactive systems wait for instability before acting. Predictive systems analyze trends and intervene early.

Integration of IoT for Real-Time monitoring

The use of IoT-enabled sensors allows continuous visibility into grid conditions. Real-time monitoring improves situational awareness and ensures that shedding decisions are based on current data rather than assumptions.

2.2.1.3 Weaknesses

High Infrastructure and Deployment Cost- In developing regions where smart grid infrastructure is limited, the cost of deployment may be prohibitive. This reduces practical applicability in areas that suffer most from frequent load shedding.

Heavy Dependence on Historical Data- Machine learning models require large, high-quality datasets for training. In regions where digital load records are incomplete or unavailable, the model may produce inaccurate predictions

2.2.2. IoT-Based Load Management with Priority Load Control [4]

Ghosh and Mitra designed a hierarchical, priority-based load shedding system that relies on IoT-enabled devices for distributed control. Their methodology involved a Sensing Layer- IoT-enabled smart plugs or relay modules were used to monitor individual appliances. Sensors measured real-time electrical parameters including voltage, current, and instantaneous power

consumption, a Communication Layer that collected data was transmitted via Wi-Fi or GSM modules to a local microcontroller, a control layer and a user interface layer.

No historical data was required; the system relied on real-time measurements for immediate decision-making.

The core methodology involved a rule-based algorithm that measure total current load and compare measured load with the predefined maximum capacity threshold.

If the threshold is exceeded it identifies low-priority loads from the priority database and then disconnect loads until the total demand is below the safe threshold.

The algorithm was simple and implemented directly on the microcontroller, minimizing computational requirements.

2.2.2.1 Key Findings

Protection of Critical Loads during Overload Conditions- One of the findings of Ghosh and Mitra's study was that hierarchical prioritization substantially improved the protection of essential appliances during peak demand events. Unlike traditional feeder-based shedding, where entire areas are disconnected.

Decision-making occurred locally within IoT-enabled controllers rather than centralized servers thus the system demonstrated fast response times. During simulated overload scenarios, the system disconnected lower-priority loads almost immediately.

2.2.2.2 Strengths

User Empowerment and Customization- This system is user-centered design. Users are no longer passive recipients of load shedding but active participants.

Transparent Decision Logic

The hierarchical shedding model is easy to understand and implement. Its decision rule is lower priority loads are shed first, while higher priority loads are preserved.

Reduced Economic and Social Impact- By ensuring that critical loads remain active, the system minimizes business interruption losses and food spoilage.

2.2.2.3 Weaknesses

Static Prioritization- load priorities are fixed unless manually changed. This creates rigidity. An appliance marked "low priority" remains low priority even if circumstances change. This lack of dynamic adaptation limits optimization efficiency.

Absence of Predictive Intelligence- The system operates reactively. It disconnects loads only after demand exceeds supply. Thereby overload conditions may escalate before corrective action

No Programmable Time Scheduling- The framework lacks the ability to schedule shedding during predefined time windows. This limits temporal flexibility.

Limited Context Awareness- prioritization remains manually controlled rather than intelligently adaptive.

2.2.3 Programmable Load Shedding Schedules in Smart Grids [5]

The project proposed a user-programmable load shedding system with an emphasis on cost and time optimization. The methodology consists of several interconnected components;

A mobile application allowed users to define shedding schedules. Users could input preferred time windows, maximum allowable durations, and tariff constraints.

An Optimization Layer was employed to compute optimal shedding periods. Constraints considered included: Peak load times, tariff pricing and user-defined schedules

A Control Layer automated relay controllers executed the shedding commands according to the calculated schedule.

Data Collection- The methodology relies mainly on user-input schedules and historical load patterns. No real-time predictive analytics were performed; the system assumes that load patterns and peak periods are known in advance.

Optimization Algorithm- The process involves defining all possible shedding windows based on user input and grid constraints.

The algorithm is static once executed; it does not adapt to unplanned grid events.

2.2.3.1 Key findings

The study demonstrated approximately 22% reduction in peak demand by shifting or shedding loads during high-consumption periods by aligning shedding schedules with peak hours, the system contributed to improved demand shaping.

Second finding was the reduction in consumer electricity bills because users could program shedding during high-tariff periods, the system optimized energy usage economically. This introduces a financial incentive for participation in demand-side management.

2.2.3.2 Strengths

High flexibility through programmability- Users can define specific shedding hours and constraints and preference.

Encourages energy awareness- By exposing users to scheduling tools, the system promotes conscious energy consumption behavior.

2.2.3.3 Weaknesses

Lack of real-time adaptive reconfiguration- Once a schedule is created, it remains fixed unless manually modified. If unexpected grid stress occurs outside scheduled times, the system may fail to respond optimally.

No predictive forecasting integration- The system does not forecast demand patterns dynamically. It relies on user-defined assumptions rather than data-driven prediction.

Absence of intelligent load prioritization- The system treats loads generically during scheduled shedding periods. It does not dynamically evaluate appliance importance.

2.3 Knowledge gap

Despite the significant advancements demonstrated in the reviewed studies, an examination of the literature reveals substantial conceptual gaps that remain unresolved in the field of intelligent load shedding systems.

The study by Singh et al. (2021) emphasizes predictive analytics through machine learning integration. While this approach improves grid stability by anticipating overload conditions, it is primarily utility-centered and lacks a consumer-oriented approach.

The work by Ghosh and Mitra (2022) introduces priority-based load control, addressing fairness and protection of critical appliances. However, the prioritization mechanism remains static. Loads are assigned fixed categories that do not dynamically adjust based on contextual factors such as time of day, occupancy, economic considerations, or evolving grid conditions.

Elshaer and Mostafa (2023) address programmability through time-based scheduling and cost optimization. Their system empowers users to define shedding windows aligned with personal preferences. However, once schedules are established, they remain fixed unless manually reconfigured.

Some common gaps we identified include:

Adaptive mechanisms are insufficient. In highly volatile grid environments, static schedules or fixed priority assignments may not adequately respond to real-time disturbances. A

dynamic feedback mechanism capable of continuous learning and adjustment is largely missing in current implementations.

Applicability in developing regions is inadequately addressed. Many predictive models assume advanced digital infrastructure and large-scale data availability, conditions that may not exist in areas frequently affected by load shedding. There is a need for a system that balances intelligence with practical deploy ability.

2.4 How our proposed system differs from existing works and the gap it fills

Practical deploy ability in infrastructure-limited Regions- Many predictive systems assume advanced digital infrastructure and extensive historical datasets. The proposed design balances intelligence with simplicity, allowing deployment in regions with limited smart grid penetration. The architecture is scalable and adaptable to both residential and small commercial environments.

The proposed system fills this gap by offering a practical solution tailored for environments experiencing frequent load shedding.

2.5 IoT Based Load Shedding Time Management System

The IoT-based system supports adaptive and real-time adjustments to load shedding schedules. If demand decreases or additional power becomes available, the system can automatically reduce shedding duration or restore power earlier than planned. Conversely, during sudden demand spikes or system faults, the system can instantly initiate shedding actions without waiting for manual commands. This intelligent time management capability improves grid stability, reduces downtime, enhances operational efficiency, and significantly increases customer satisfaction compared to conventional load shedding techniques.

2.6 Programmable Interface

The programmable interface is an important component of the proposed IoT-based load shedding system, as it provides a flexible and interactive platform through which system operators can monitor, configure, and control load shedding operations. The interface serves as the main point of interaction between the human operator and the automated control system, enabling informed decision-making and dynamic system management.

Through the programmable interface, operators are able to define load shedding policies, scheduling rules, and priority levels for different consumer categories.

CHAPTER THREE

3. MATERIALS AND METHODS

3.1 Introduction

In this proposed system, we have both hardware and software components. The hardware consists of most of the components mentioned in the previous chapter. In this chapter, a purposive study on the design of the load shedding time management system with a programmable interface will be explained in details. The components used in this project are; LED, capacitors, resistors, voltage sensor, current sensor, microcontrollers as in the table 3.1.

Figure 3.0 shows a detailed the system model connection on tinkercad as modeled by us.

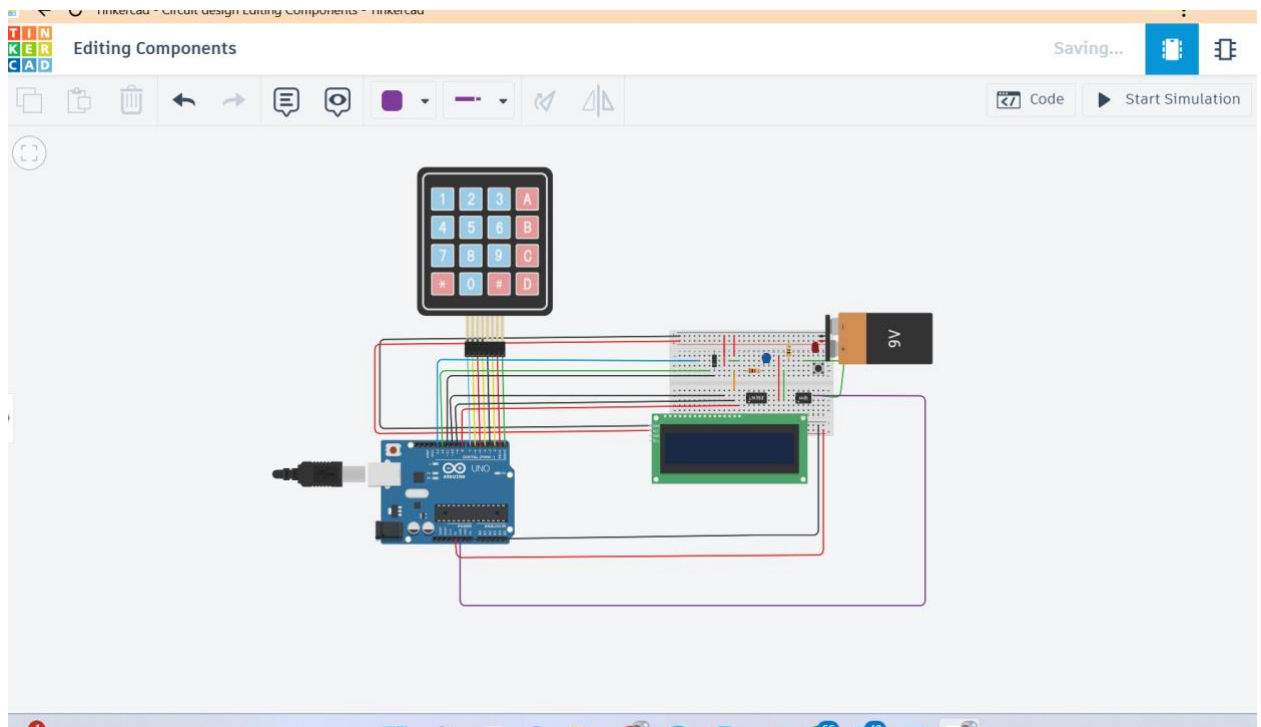


Figure 3.0 system model demonstration

The system design and functional mechanism is explained thoroughly up to point where load shedding occurs in steps from heading 3.4

3.2 Hardware Specifications

Below is a table listing the hardware specifications of the components I will require to complete the project.

Table 3.I Shows materials and their specifications

TABLE I
HARDWARE SPECIFICATIONS

NO.	MATERIAL	SPECIFICATION
1	Microcontroller	ESP32
2	Display	7 Segment
3	Resistors	300 Ω , 4.7k Ω
4	Capacitors	1000 μ F
5	Cables	1.5mm ²
6	Breadboard	850 Tie-points
7	Transformer	12VAC, 2A
8	Adapter	12V DC
9	Push button	Tactile push-button
10	Switch	Normal
11	Load	50W bulb
12	IC	LM7805, ULN 203
13	Diodes	1N4007
14	IC Sockets	16 pin

3.2.1 Microcontroller

Figure 3.1 shows an Arduino microcontroller [2]. It acts as the central control unit of the entire system. In this project, the microcontroller is responsible for acquiring data from sensors, executing the load shedding algorithms, handling communication with the IoT platform, managing timing and scheduling operations, and generating control signals to drive the relay or switching circuits. It enables real-time decision-making and ensures that load shedding actions

are carried out automatically based on system conditions and programmed rules. The esp32 microcontroller is chosen. It has a high processing speed and low power consumption.

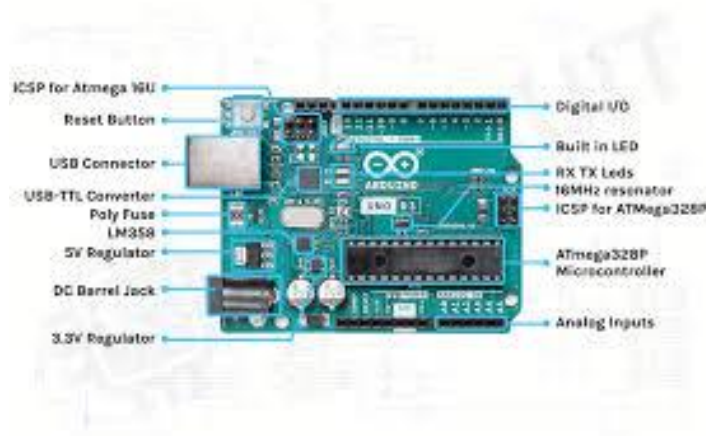


Figure 3.1 An Arduino microcontroller

3.2.2 LCD Display

Figure 3.2 shows LCD display [2]. The LCD display serves as a human-machine interface for the system. It provides visual feedback to the user by displaying important information such as current system status, power availability, load conditions, operational mode (automatic or manual), and load shedding events. In this project, the display improves usability by allowing operators to monitor system performance locally without relying solely on remote dashboards.

We're using the 16x2 LCD display. It reduces pin usage and displays load shedding time.



Figure 3.2 LCD display

3.2.3 Resistors

Figure 3.3 shows Resistors [2]. Resistors are passive electronic components that oppose the flow of current in a circuit. Their primary function is to control current levels and divide voltage as required. In this project, resistors are used for signal conditioning, current limiting to protect the microcontroller's input/output pins, biasing electronic components, and ensuring correct voltage levels for sensors and interface devices. They also help in pull-up and pull-down configurations for digital inputs such as buttons and switches. We're using 300Ω for LCD current limiting and $4.7k\Omega$ for pull-up resistors. They limit current to 10-20mA and stabilize digital inputs respectively.

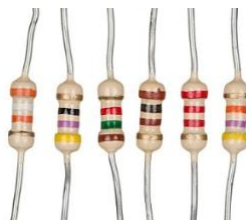


Figure 3.3 Resistors

3.2.4 Capacitors

Figure 3.4 shows a Capacitor [2]. Capacitors store electrical charge and release it when needed. They are widely used for filtering, decoupling, and energy storage. In this project, capacitors are used to smooth out fluctuations in the power supply, reduce electrical noise, stabilize voltage levels, and prevent sudden voltage drops that could reset or damage sensitive components. Filter capacitors we'll use 1000 μ F. It has a smooth DC.



Figure 3.4 Capacitor

3.2.5 Connectors

Figure 3.5 shows Connectors [2]. These are conductive wires that provide physical electrical connections between components. In this project, cables are used to interconnect the microcontroller, sensors, display, relay modules, and power supply. They facilitate both power distribution and data transmission, ensuring proper communication and signal flow throughout the system. We'll use PVC copper insulated 1.5mm². It handles up to 15A safely.

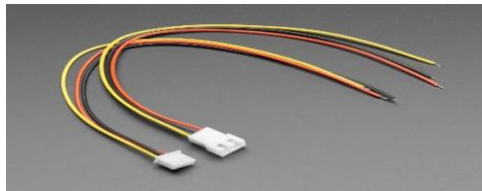


Figure 3.5 Connectors

3.2.6 Breadboard

Figure 3.6 shows a Breadboard [2]. A breadboard is a reusable prototyping platform that allows electronic components to be connected without soldering. In this project, the breadboard is used during the design and testing phase to quickly assemble circuits, experiment with different configurations, and debug the system before final deployment on a printed circuit board (PCB). We'll use 830 Tie-points. It is easy to debug and no soldering required.



Figure 3.6 Breadboard

3.2.7 Transformer

Figure 3.7 shows a Transformer [2]. A transformer is an electrical device used to change voltage levels in an AC power system. In this project, the transformer steps down the high mains voltage to a lower and safer voltage suitable for the control circuitry. This ensures electrical safety and protects sensitive electronic components from high voltage damage.

We'll use 12VAC rated step-down transformer. It improves stability and prevents overheating.



Figure 3.7 Transformer

3.2.8 Adapter

The adapter is a power conversion unit that converts AC voltage from the mains into regulated DC voltage required by the electronic components. In this project, the adapter supplies stable DC power to the microcontroller, sensors, and communication modules, enabling continuous and reliable system operation. We'll use 12V DC adapter. It's already regulated and small and efficient.

3.2.9 Load

The load represents the electrical appliances or systems being controlled, such as lamps, fans, motors, or simulated power lines. In this project, the load is the target of the load shedding process, meaning it is the component that is disconnected or reconnected based on system

decisions. The load models real-world consumers in a power distribution system. We'll use a 50W bulb. It saves on power.

3.2.10 Switch

A switch is a mechanical device used to open or close an electrical circuit. In this project, switches are used to enable or disable automatic load shedding, select operating modes, or manually control certain system functions. They provide a basic but important means of system control and override.

3.2.11 Integrated Circuit

Figure 3.8 shows an IC [2]. An integrated circuit is a miniaturized electronic circuit that performs specific functions such as signal processing, communication, power regulation, or logic control. In this project, ICs may include voltage regulators, communication chips, or driver circuits that support the operation of the microcontroller and other modules. We chose LM7805 and ULN203. They protect MCU and amplifies current to drive relays.



Fig 3.8 IC

3.2.12 Diodes

Figure 3.9 shows a Diode [2]. Diodes are semiconductor devices that allow current to flow in only one direction. In this project, diodes are used for rectification, reverse polarity protection, and protection against voltage spikes caused by inductive loads such as relays. They help prevent damage to components and improve system safety. We'll use 1N4007 rated 1A. It prevents back-EMF damage to MCU and provides rectifier protection.



Fig 3.9 Diode

3.2.13 Sensors

A sensor is a device that detects and responds to changes in a given environment. It converts physical phenomenon such as temperature, light, motion or pressure into electrical signals that can be measured and recorded. Sensors are used in a wide range of applications from smartphones, industrial machinery to home security systems. They play a very key role in functioning of modern technology by enabling systems to monitor, analyze and respond to different conditions.

3.2.14 Voltage sensors

Figure 3.10 shows a Voltage Sensor [3]. A voltage sensor is a device that measures the amount of voltage in an electrical circuit. It converts the voltage level into a readable signal, either an analog or digital output which can be monitored and recorded by other devices or systems. They are used in various applications like in power management systems, battery monitoring and in renewable energy systems to ensure proper voltage levels and maintain the stability and safety of electrical systems. We'll use voltage range 0-230VAC. Safe isolation from mains and suitable for 230V systems.



Figure 3.10 Voltage sensor

3.2.15 Current sensor

A current sensor is a device that detects and measures the flow of electric current in a circuit. It converts this measurement into a readable signal, either an analog or digital output, which can be monitored and processed by other systems. They are used in various applications such as power management, battery monitoring, industrial automation and motor control. They ensure there is proper current levels, protection against overcurrent conditions and optimizing systems' performance. ACS712 current range +/- 30A. Has a safety margin and suits typical domestic loads.

3.3 Software Specifications

An Integrated Development Environment (IDE) that is compatible with the microcontrollers will be used for coding and uploading the codes to the microcontroller boards. Arduino IDE will be used for this project as it can be used for both Arduino Uno and 8052.

3.4 System Specification and Functional Design

Figure 3.11 shows component Block Diagram [2]. In this step, appropriate hardware and software materials are selected. Hardware components include the microcontroller, current and voltage sensors, relay modules, display unit, communication modules, power supply units, resistors, capacitors, diodes, ICs, breadboard, and connecting cables. Software materials include programming environments such as Arduino IDE, IoT platforms, and database services [3]. These materials provide the technical foundation for system development.

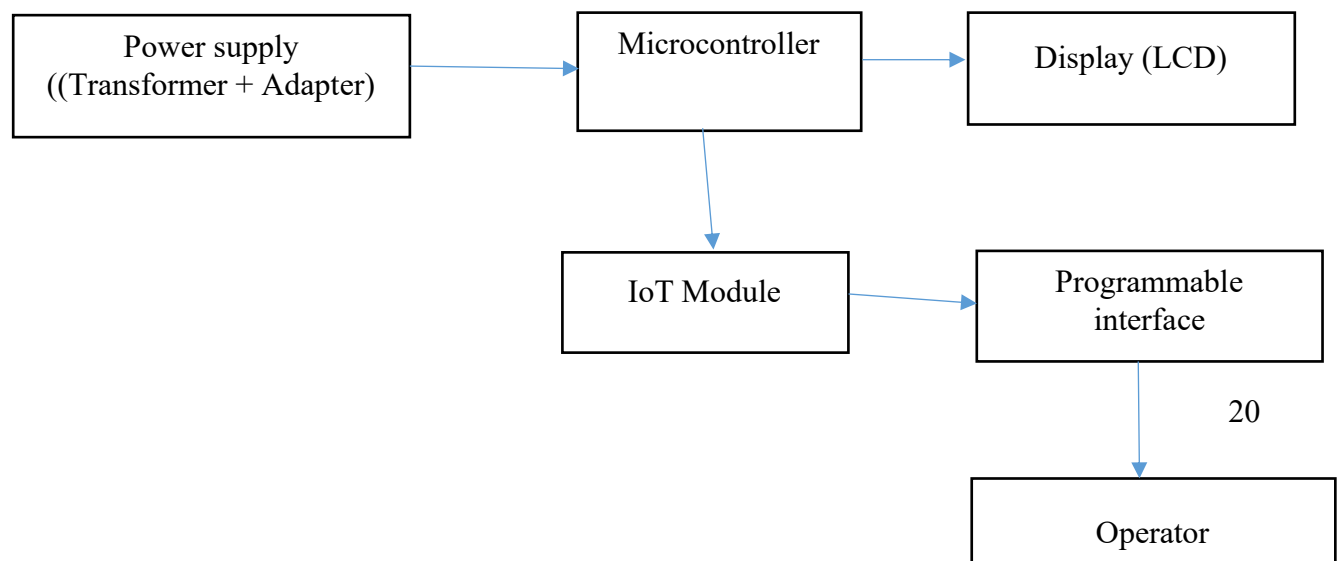


Figure 3.11 block diagram

3.5 Power Supply Design Using Transformer and Adapter

The power subsystem is designed using a step-down transformer to convert high AC mains voltage to a lower AC voltage. An AC–DC adapter is then used to rectify and regulate this voltage into stable DC levels required by the microcontroller and ICs [7]. Capacitors are added for filtering and smoothing, while diodes provide rectification and reverse polarity protection. This ensures a clean and stable power source for the embedded system.

3.6 Control Unit Implementation Using Microcontroller and ICs

Figure 3.12 shows layout Demonstration [2]. The microcontroller is configured as the main processing unit. Supporting ICs, such as voltage regulators and driver ICs, are connected through IC sockets for easy replacement and debugging. Resistors are used for current limiting, pull-up and pull-down configurations, and voltage division. This stage establishes the embedded control environment.

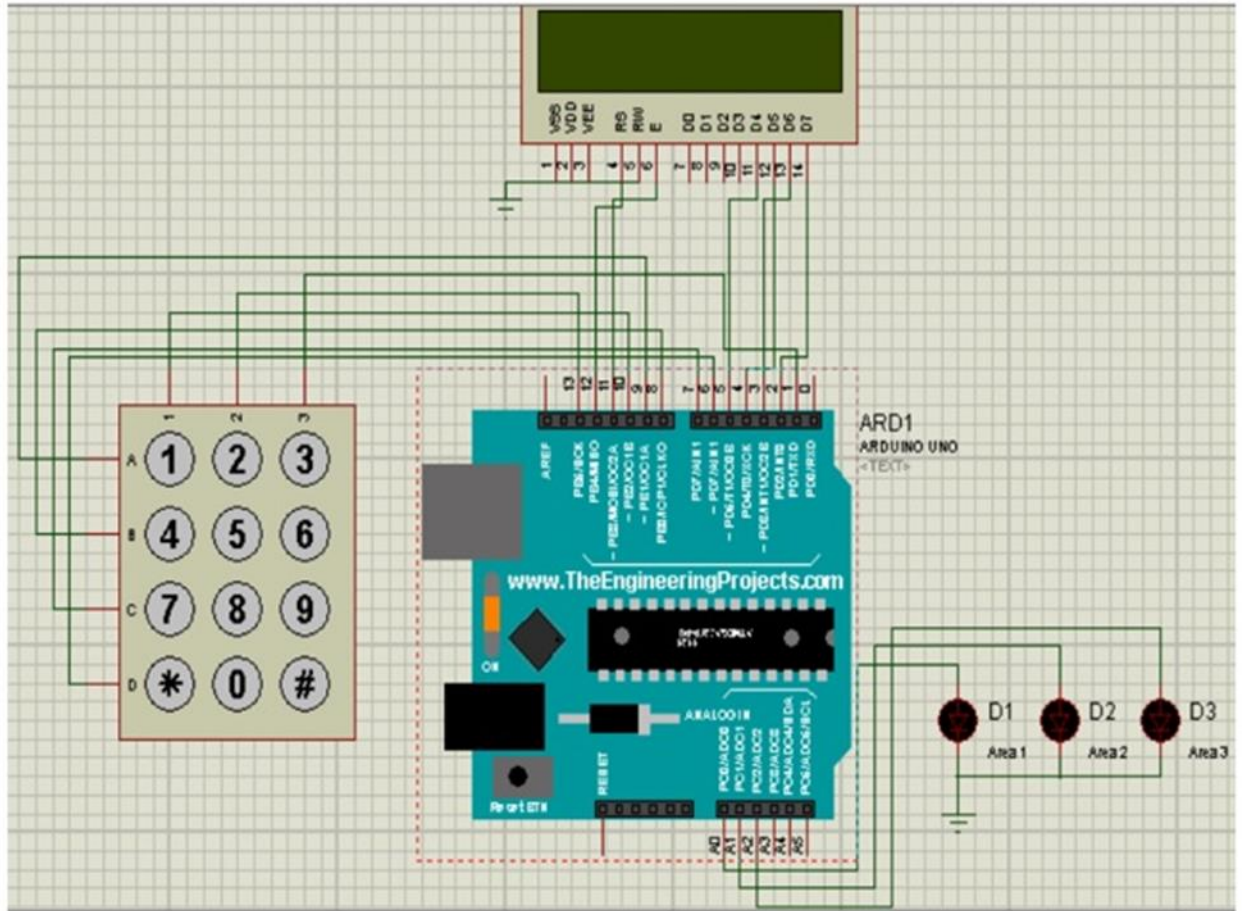


Figure 3.12 demonstration

3.7 Circuit Prototyping on Breadboard and Cabling

All components are assembled on a breadboard using connecting cables. This allows rapid prototyping and testing of the system without soldering [8]. The microcontroller, display, switches, push buttons, and relay modules are interconnected according to the circuit design.

3.8 Input Interface Using Push Buttons and Switches

Push buttons are integrated to provide manual input functions such as system reset, mode selection, and test activation. Switches are used to toggle between automatic and manual control modes [9]. Resistors are configured in pull-up or pull-down mode to ensure stable digital input signals.

3.9 Output Interface Using Display Module

The display module is interfaced with the microcontroller using digital communication protocols. It displays real-time system information such as system status, load state, time scheduling parameters, and load shedding events. This forms the local human–machine interface [10].

3.10 Load Control Using Relays and Loads

The relay module acts as an electrically controlled switch. The microcontroller sends logic signals to the relay driver circuit, which physically connects or disconnects the load. This implements the actual load shedding process. Diodes are connected across relay coils to suppress back electromotive force (EMF) and protect the microcontroller.

3.11 IoT Communication and Programmable Interface

The microcontroller communicates with the IoT platform using a communication module. Real-time system data is transmitted to the cloud, where the programmable interface allows operators to configure time schedules, priority levels, and load shedding policies. These parameters are sent back to the microcontroller for execution.

3.12 Load Shedding Algorithm Implementation

The control algorithm is programmed into the microcontroller. It continuously monitors system parameters and compares them with threshold values [11]. When overload or peak conditions are detected, the algorithm triggers relay switching based on priority and time scheduling logic defined in the programmable interface.

3.13 Testing, Calibration and Validation

The complete system is tested under different operating conditions. Loads are varied to simulate peak and normal demand. Sensor accuracy, relay response time, power stability, and IoT communication reliability are evaluated. Capacitors and resistors are adjusted where necessary to improve signal stability and timing accuracy.

3.14 Final Integration and Demonstration

After successful testing, the system is integrated into a final setup. The breadboard prototype is replaced with a more stable configuration if required. The system is demonstrated by showing real-time monitoring on the display, programmable scheduling through the interface, and automated load shedding through relay control.

CHAPTER FOUR

4. Expected results and discussion

4.1 Introduction

In this chapter we will discuss the anticipated results of load shedding time management system with a programmable interface. We will discuss the expected performance of the model, potential challenges, and the implications of the results on future network security technologies.

The primary focus is on the system's effectiveness to be a fully functional IoT-based load shedding time management system capable of intelligent, scheduled, and priority-based power control with remote programmability, real-time monitoring, and reliable hardware implementation.

4.2 EXPECTED RESULTS

Stable and Reliable Power Regulation

The completed system is expected to provide stable and reliable power regulation through the use of the transformer, adapter, voltage regulator ICs, capacitors, and diodes. The transformer will step down the AC mains voltage to a safer level, while the adapter and regulation stage will convert it into stable DC suitable for the microcontroller and other components. Capacitors will smooth voltage ripple, and diodes will provide reverse polarity and rectification protection. As a result, the system should operate without voltage fluctuations, random resets, or component damage, ensuring consistent and uninterrupted performance.

Accurate Load Monitoring and Control

The microcontroller is expected to continuously monitor the system and execute control decisions based on programmed logic. It should detect overload conditions or threshold violations in real time and respond immediately by activating relay switching mechanisms. When abnormal conditions occur, the system should disconnect selected loads to prevent system

failure. Once the load returns to safe operating conditions, the controller should restore power appropriately. This will ensure fast response time, improved system stability, and efficient load management.

Effective Time-Based Load Scheduling

The time management functionality is expected to allow automatic scheduling of load shedding operations according to predefined time intervals. During peak hours, non-essential loads should be disconnected, while during off-peak periods, power should be restored. The system should follow programmed schedules accurately without manual intervention. This will help reduce peak demand, distribute energy more efficiently, and improve overall power utilization.

Prioritized Load Management

The system is expected to classify loads into different priority levels, ensuring that critical loads remain operational while less important loads are shed first. In situations where demand exceeds supply, the controller should disconnect low-priority loads in a structured and logical sequence. This intelligent prioritization will reduce inconvenience compared to total shutdowns and ensure that essential services remain uninterrupted.

Functional IoT Communication

The IoT module integrated into the system is expected to enable real-time communication between the hardware unit and a cloud-based server. It should successfully transmit system status data and receive control commands remotely. Reliable network connectivity will allow operators to monitor load conditions, modify schedules, and implement control actions from remote locations. This feature will enhance flexibility, accessibility, and system responsiveness.

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